



Climate.Og.ai Dashboard

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ØG AI Inference Energy & Carbon Transparency Plan

Decentralized AI Operating System + Phala TEE Verification + Blockchain Aggregation

Executive Summary

The AI industry operates on **black box infrastructure** — no visibility into energy consumption per request, no cryptographic proof of claims, and no public audit trail.

ØG's decentralized AI operating system (dAIOS) can lead the industry by becoming the **first AI platform to deliver verifiable, per-inference energy data** — an achievement no centralized AI provider can match.

Three-Pillar Approach:

1. 🇮🇹 **Real-Time Measurement** - Capture energy usage for every inference via Service SDK
2. 🏠 **Cryptographic Verification** - Verify via Phala's GPU Trusted Execution Environments (TEEs)
3. 🌐 **Public Transparency** - Aggregate on-chain and display via climate.Og.ai

Strategic Value & Business Impact

Market Differentiation

- 🏆 **First-mover advantage** - Only platform with cryptographically verified, per-inference energy data
- 🏰 **Unique competitive moat** - Hardware-attested results that centralized AI cannot replicate
- 📈 **Network effects** - Providers compete for 'Verified' status, driving adoption

User Benefits

- ✅ **Trustworthy metrics** for ESG compliance and carbon accounting
- ✅ **Provider selection** based on verified efficiency data
- ✅ **Wallet-level tracking** for organizational sustainability reporting

Technical Advantages

- ✅ **Full network coverage** - Every inference measured from genesis
- ✅ **Model-specific accuracy** - Granular data per model and provider
- ✅ **Immutable audit trail** - On-chain aggregation ensures transparency

Competitive Analysis

Challenge	Centralized AI (OpenAI, Anthropic, Google)	ØG Opportunity
Per-inference energy transparency	❌ None	✅ Network-wide
Cryptographic proof	❌ None	✅ Via Phala TEE
Public audit trail	❌ None	✅ On-chain from genesis
Provider-specific metrics	❌ None	✅ Full granularity
Verifiable sustainability	❌ Marketing only	✅ Hardware-attested

Implementation Roadmap

Phase	Stage	Technical Implementation	Deliverables
Phase 1	Testnet	Real-Time Energy Measurement	• Service SDK enhancement for energy headers: - x-gpu-wh: <float> (watt-hours) - x-gpu-avg-watts + x-gpu-active-ms • Service metadata extension: - energyPerToken field - carbonIntensity field • Broker middleware labels data: - Verified (Phala TEE) - Reported (self-reported) - Estimated (model profile)
Phase 2	Mainnet Launch	Phala TEE Verification	• Providers run inference in GPU TEEs • Each request includes RA report: - Model ID/version - Input/output hashes - Measured Wh • SDK processResponse() validates attestation • 'Verified' badge on dashboard
Phase 3	Full Production	Public Dashboard & Aggregation	• On-chain storage by wallet/model/provider • climate.0g.ai displays: - Provider rankings - Historical trends - Carbon footprints • Public API endpoints

Technical Architecture

Data Flow

1. **Service SDK** → Providers add energy headers to inference responses
2. **Broker Middleware** → Captures, verifies, and labels energy data via `processResponse()`
3. **ØG Chain** → Aggregates Wh and CO₂ metrics on-chain
4. **Climate Dashboard** → Displays real-time and historical data

SDK Integration Points

- **Service Metadata** → Extend to include energy and carbon fields

- **Response Processing** → Enhanced `processResponse()` to capture energy headers
- **TEE Verification** → Leverage TeeML infrastructure for energy attestation

Verification Levels

-  **Verified (Phala TEE)** - Hardware-attested, tamper-proof measurements
 -  **Reported** - Self-reported data without cryptographic proof
 -  **Estimated** - Derived from model profiles and token counts
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Resource Requirements

Joint Implementation (ØG + ecoBridge Collaboration):

- Extend Service SDK to support energy tracking headers and metadata
 - Update SDK documentation for `x-gpu-wh` headers implementation
 - Implement broker middleware for energy data capture and verification
 - Develop `processResponse()` enhancements for energy attestation
 - Create on-chain aggregation infrastructure for Wh/CO₂ metrics
 - Design API endpoints for energy data access
 - Coordinate with third-party provider for MiCA compliance requirements
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Strategic Outcome

ØG becomes the only AI platform where:

Every inference is measurable — Complete network visibility from genesis

Select inferences are cryptographically verified — Hardware-attested proof via Phala TEE

The entire history is publicly auditable — Immutable on-chain records

This creates unprecedented transparency that centralized AI cannot match.

Next Steps

1. **ØG approval** of implementation plan and resource allocation
 2. **Technical specification review** - Finalize Service SDK modifications and API formats
 3. **Integration kickoff** - Align on broker middleware and TEE attestation process
 4. **Testnet deployment** - Begin Phase 1 with energy measurement implementation
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This positions ØG as the definitive leader in transparent, sustainable AI infrastructure.

Let's build the future of verifiable AI sustainability together.
